

FRUITS

A **fruit** is the ripened (mature) ovary of a flower formed after fertilization.

It may also include other floral parts (called **accessory fruit** in some cases like apple).



Classification of Fruits

Fruits are mainly classified into:

1. **Simple fruits**
2. **Aggregate fruits**
3. **Multiple fruits**

Simple Fruits

Develop from **one ovary of a single flower**.

A. Dry Simple Fruits

Pericarp (fruit wall) becomes **dry and hard**.

Dry Simple Fruits

These are **simple fruits** in which the pericarp becomes **dry at maturity**.

They are divided into 3 main types:

1. Indehiscent
2. Dehiscent
3. Schizocarpic

1 Indehiscent Dry Fruits

Do not split open at maturity.

Fruits in which seeds remain enclosed and are dispersed along with the fruit.

Types of Indehiscent Fruits

i. Achene

- Small, one-seeded fruit

- Seed is free from fruit wall

Examples:

- Sunflower
- Buttercup

ii. Caryopsis (Grain)

- Seed coat is fused with fruit wall

Examples:

- Wheat
- Rice

iii. Nut

- Hard, woody fruit
- Usually one-seeded

Examples:

- Chestnut
- Acorn

iv. Samara

- Wing-like structure helps in wind dispersal

Examples:

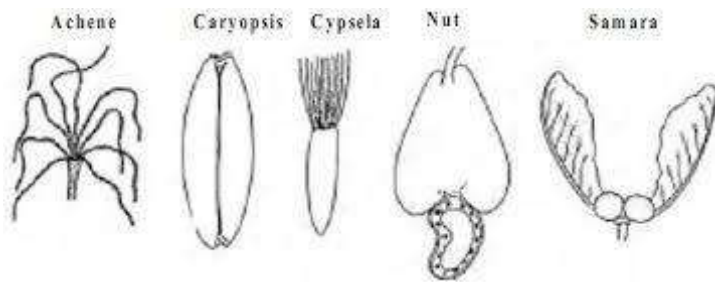
- Maple
- Elm

v. Cypsela

- Similar to achene but develops from inferior ovary

Examples:

- Dandelion
- Sunflower (also considered here in some cases)



2 Dehiscent Dry Fruits

Split open at maturity to release seeds.

Fruits that burst along definite lines called **sutures**.

Types of Dehiscent Fruits

i. Legume

- Splits along two sides

Examples:

- Pea
- Bean

ii. Follicle

- Splits along one side only

Examples:

- Calotropis
- Magnolia

iii. Capsule

- Splits in different ways

Examples:

- Cotton
- Datura

Types of capsule:

- **Loculicidal** (splits along locule)

- **Septicidal** (splits along septa)
- **Poricidal** (pores open, e.g., poppy)

iv. Siliqua

- Long fruit, splits into two valves

Examples:

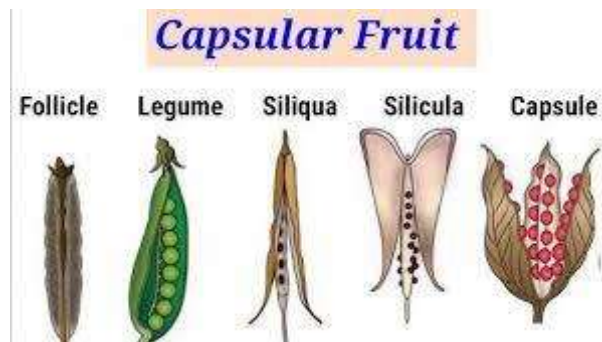
- Mustard
- Radish

v. Silicula

- Short and broad version of siliqua

Examples:

- Shepherd's purse



3 Schizocarpic Fruits

Show features of both indehiscent and dehiscent fruits.

Fruits that split into **one-seeded segments (mericarps)** but do not open to release seeds.

Types of Schizocarpic Fruits

i. Lomentum

- Breaks into one-seeded pieces

Examples:

- Acacia
- Mimosa

ii. Cremocarp

- Splits into two mericarps

Examples:

- Coriander
- Fennel

iii. Regma

- Splits into three or more parts

Examples:

- Castor
- Euphorbia

iv. Carcerulus

- Splits into four one-seeded parts

Examples:

- Ocimum (Tulsi)

B. Simple fleshy fruit

simple fleshy fruit is formed from a single carpel or fused carpels of one flower, where the pericarp becomes soft and edible when ripe.

Types of Simple Fleshy Fruits

1. Berry

- Entire pericarp is soft and juicy.
- Seeds are embedded in the pulp.
- **Examples:**
- Tomato
- Grape
- Banana

2. Drupe (Stone Fruit)

- Has three layers:
- **Exocarp** (outer skin)
- **Mesocarp** (fleshy middle)
- **Endocarp** (hard inner stone enclosing seed)
- **Examples:**
- Mango
- Peach
- Coconut

3. Pome

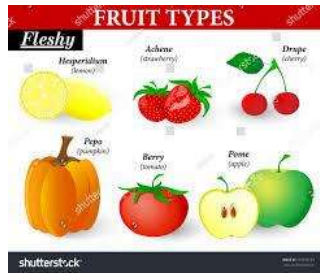
- Fleshy part develops mainly from the **thalamus (receptacle)**, not just the ovary.
- The ovary forms a **core** containing seeds.
- **Examples:**
- Apple
- Pear

4. Hesperidium

- A special type of berry with a **leathery rind** and **juicy segments** inside.
- **Examples:**
- Orange
- Lemon

5. Pepo

- Develops from an **inferior ovary**.
- Has a **hard outer rind** and soft inner pulp.
- **Examples:**
- Watermelon
- Pumpkin
- Cucumber



2. Aggregate fruit

An **aggregate fruit** develops from a **single flower that has many free carpels (apocarpous gynoecium)**. Each carpel forms a small fruitlet, and all the fruitlets together form one aggregate fruit.

- Originates from **one flower**
- Has **many separate carpels**
- Each carpel develops into a **small fruitlet**
- Fruitlets are **grouped together on a single receptacle**

Types of Aggregate Fruits

Aggregate fruits are classified based on the type of individual fruitlets:

1. Etaerio of Achenes

- Fruitlets are small, dry, one-seeded achenes

Examples:

- Strawberry

2. Etaerio of Follicles

- Each fruitlet is a follicle (splits along one side)

Examples:

- Magnolia

3. Etaerio of Drupes (Drupelets)

- Each fruitlet is a small drupe

Examples:

- Raspberry

- Blackberry

4. Etaerio of Berries

- Each fruitlet is a small berry

Examples:

- Custard apple (Sitaphal)



•



•

Multiple Fruit

A **multiple fruit** (also called a **composite fruit**) develops from the **entire inflorescence (cluster of flowers)** rather than a single flower. In this case, the ovaries of many flowers along with other floral parts **fuse together** to form one large fruit.

- Formed from **many flowers (inflorescence)**
- Individual fruits **merge into a single mass**
- Often includes **ovaries + floral parts** (like bracts, axis)

Types of Multiple Fruits

1. Sorosis

- Develops from a **spike, spadix, or catkin type inflorescence**
- Fruits fuse along with the **axis and other floral parts**, making a fleshy mass

Examples:

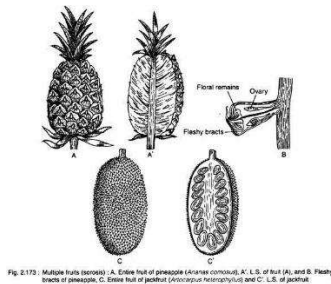
- Pineapple
- Jackfruit
- Mulberry

2. Syconus

- Develops from a **hypanthodium inflorescence** (a hollow, fleshy structure with flowers inside)
- The fruit forms a **fleshy, hollow receptacle** with multiple small fruits inside

Examples:

- Fig



Morphology of Mushroom

A **mushroom** is the fleshy, spore-bearing fruiting body of certain fungi, mainly belonging to the group **Basidiomycetes**. The vegetative part is called **mycelium**, which grows underground or within organic matter.

Main Parts of a Mushroom

1. Mycelium

- Network of thread-like **hyphae**
- Found in soil, wood, or organic material
- Responsible for nutrient absorption

2. Stipe (Stalk)

- Cylindrical structure that supports the cap
- Elevates the cap for better spore dispersal

3. Pileus (Cap)

- Umbrella-shaped top portion
- Protects the reproductive structures underneath

4. Gills (Lamellae)

- Thin plates located under the cap
- Bear **basidia**, which produce spores

5. **Annulus (Ring)**

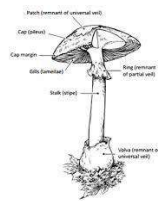
- Ring-like structure on the stipe
- Remnant of the partial veil

6. **Volva**

- Cup-like structure at the base
- Remnant of the universal veil

7. **Spores**

- Reproductive units
- Produced in basidia and dispersed by wind



○

Economic Importance of Mushrooms

Beneficial Importance

1. Food Source

- Rich in proteins, vitamins (B-complex), and minerals
- Low in fat and calories

2. Medicinal Value

- Some species have **antibiotic, anticancer, and immune-boosting properties**
- Example: *Ganoderma lucidum*

3. Decomposition

- Help in **recycling nutrients** by decomposing organic matter

4. Biotechnology

- Used in enzyme production and research

5. Income Generation

- Mushroom farming is a profitable small-scale industry

Harmful Importance

1. Poisonous Species

- Some mushrooms are highly toxic and can cause death

2. Spoilage

- Can damage stored food and crops

3. Allergies

- Spores may cause respiratory allergies in some people

Edible Mushrooms (Scientific Name + Common Name)

1. *Agaricus bisporus* — Button mushroom
2. *Pleurotus ostreatus* — Oyster mushroom
3. *Lentinula edodes* — Shiitake mushroom
4. *Volvariella volvacea* — Paddy straw mushroom
5. *Morchella esculenta* — Morel mushroom
6. *Boletus edulis* — Porcini mushroom
7. *Flammulina velutipes* — Enoki mushroom
8. *Termitomyces heimii* — Termite mushroom
9. *Auricularia auricula-judae* — Wood ear mushroom
10. *Cantharellus cibarius* — Chanterelle

Poisonous Mushrooms (Scientific Name + Common Name)

1. *Amanita phalloides* — Death cap
2. *Amanita muscaria* — Fly agaric
3. *Amanita virosa* — Destroying angel
4. *Galerina marginata* — Deadly galerina

5. *Cortinarius rubellus* — Deadly webcap
6. *Clitocybe dealbata* — Ivory funnel
7. *Inocybe patouillardii* — Fibrecap
8. *Gyromitra esculenta* — False morel
9. *Omphalotus olearius* — Jack-o'-lantern mushroom
10. *Lepiota brunneoincarnata* — Deadly dapperling

Role of micro-organisms in biotechnology

1. Microorganisms in Fermentation Technology

Fermentation is the process where microorganisms convert sugars into useful products.

Examples:

- *Saccharomyces cerevisiae* (yeast) → produces **ethanol** (used in alcohol, biofuel)
- *Lactobacillus* → produces **lactic acid** (used in curd, yogurt)
- *Acetobacter aceti* → produces **acetic acid** (vinegar)
- **Importance:** Food production, beverages, and industrial chemicals

2. Production of Antibiotics

Some microorganisms naturally produce substances that kill or inhibit other microbes.

Examples:

- *Penicillium notatum* → produces **penicillin**
- *Streptomyces griseus* → produces **streptomycin**

Importance: Treatment of bacterial diseases

3. Genetic Engineering and Recombinant DNA Technology

Microorganisms are used to **insert and express foreign genes**.

Examples:

- *Escherichia coli* is used to produce **human insulin**
- Bacteria act as **vectors** to transfer genes

Importance: Production of hormones, vaccines, and therapeutic proteins

4. Vaccine Production

Microorganisms are used to develop vaccines that protect against diseases.

Examples:

- Attenuated or killed microbes used in vaccines
- Recombinant vaccines produced using bacteria or yeast

Importance: Disease prevention (e.g., hepatitis, polio)

5. Enzyme Production

Microorganisms produce enzymes used in industries.

Examples:

- *Aspergillus niger* → produces **citric acid & enzymes**
- *Bacillus* species → produce **proteases and amylases**

Importance: Used in detergents, food processing, textiles

6. Biofertilizers

Certain microorganisms improve soil fertility by fixing nitrogen.

Examples:

- *Rhizobium* → nitrogen fixation in legume roots
- *Azotobacter* → free-living nitrogen fixer

Importance: Eco-friendly agriculture

7. Bioremediation

Microorganisms help clean pollutants from the environment.

Examples:

- Bacteria degrade **oil spills, sewage, and toxic wastes**

Importance: Environmental protection

8. Biogas Production

Microorganisms break down organic waste anaerobically.

Examples:

- *Methanobacterium* → produces **methane gas**

Importance: Renewable energy source